Supplement to "Reporting AI-Assisted Proofs in Mathematical Risk Management"

Proofs, further results and records of the case study

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This supplement contains the material of the case study that is not needed to follow the paper: the proofs of the main results, a discussion of the role of the axioms, the version without law invariance, the version for nonnegative losses, further consequences of the representation, a summary table of the literature, and the timed record of the Lean check. References to sections, results and tables without the prefix S are to the paper. The supplement is compiled together with the paper from the same source, which is available in the repository `https://github.com/hirbod-assa/loss-ratio-lean`.

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S1 Proofs of the main results

Figure S1 shows the structure of the proof. Lemma S1.1 derives the properties (P), (N) and (CM) from the axioms ($\mathsf{P}_\exists$) and (R); Lemmas S1.1–S1.6 are specific to loss ratios; Lemmas S1.7–S1.10 reprove the representation of comonotonic additive functionals in the form needed here.

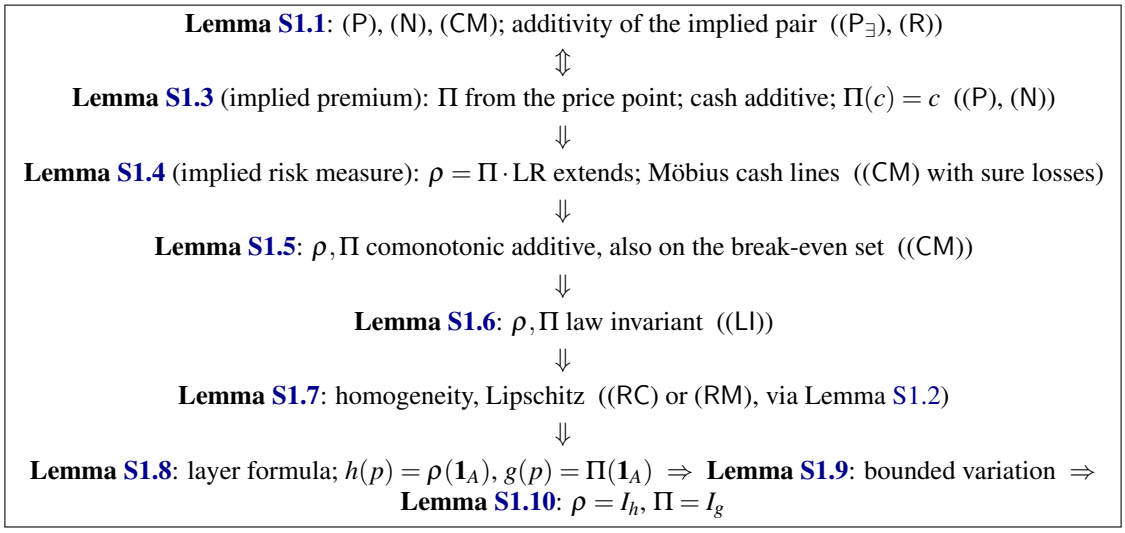


Figure S1: Logical structure of the proof of Theorems 6.1 and 6.2.

Lemma S1.1 (The derived properties)**.** *Axioms* ($\mathsf{P}_\exists$) *and* (R) *hold if and only if* (P), (N) *and* (CM) *hold. In that case, for every unpriced W (that is, $\mathrm{LR}(W) \notin \mathbb{R}$) the quantity $r(W) := s\,(\mathrm{LR}(W+s)-1)$ does not depend on $s \neq 0$, and the implied premium and the implied risk measure satisfy $\Pi(X) = -c_X^\infty$ and $\rho(X) = \Pi(X) + r(X - \Pi(X))$ and are comonotonic additive.*

Proof. Assume ($\mathsf{P}_\exists$) and (R). The quantities Π, ρ, r below are used only in the proof; they appear in no axiom.

Step 0: (P). Let $\mathrm{LR}(Z) \notin \mathbb{R}$. The book Z is comonotone with itself, so (R) applied to the pair (Z,Z) gives $\mathrm{LR}(Z+s) \in \mathbb{R}$ for every $s \neq 0$. Hence a cash line has at most one non-finite point, and by ($\mathsf{P}_\exists$) exactly one. So (P) holds; put $\Pi(X) := -c_X^\infty$, so that $\Pi(X+c) = \Pi(X) + c$ by (3).

Step 1: (N). Let p_0 be the non-finite point of the cash line of 0. Constants are comonotone, and $\mathrm{LR}(p_0) \notin \mathbb{R}$, so (R) applied to (p_0, p_0) gives $\mathrm{LR}(2p_0 + u) \in \mathbb{R}$ for all $u \neq 0$. If $p_0 \neq 0$, the choice $u = -p_0$ gives $\mathrm{LR}(p_0) \in \mathbb{R}$, a contradiction; hence $p_0 = 0$ and $\mathrm{LR}(0) \notin \mathbb{R}$. Now apply (R) to $(0,0)$ with $s = t = u$: $f(u) := u(\mathrm{LR}(u) - 1)$ satisfies $f(u) = 2f(u)$, so $f(u) = 0$ and $\mathrm{LR}(u) = 1$ for every $u \neq 0$.

Step 2: the cash-line law. Let W be unpriced. The pair $(W, 0)$ is comonotone and both books are unpriced, so (1) gives $u(\mathrm{LR}(W+u)-1) = s(\mathrm{LR}(W+s)-1) + t(\mathrm{LR}(t)-1) = s(\mathrm{LR}(W+s)-1)$ for all $s, t, u \neq 0$, by Step 1. Hence $r(W)$ does not depend on s. For any X put $X_0 := X - \Pi(X)$, which is unpriced, and $\rho(X) := \Pi(X) + r(X_0)$. For $c \neq -\Pi(X)$ write $X + c = X_0 + s$ with $s = \Pi(X) + c \neq 0$; then

$$\mathrm{LR}(X+c) = 1 + \frac{r(X_0)}{s} = \frac{\rho(X) + c}{\Pi(X) + c}.$$

In particular $\rho = \Pi \cdot \mathrm{LR}$ on D, ρ is cash additive, and $\rho(c) = \Pi(c) = c$.

Step 3: comonotonic additivity. Let X, Y be comonotone. Cash shifts preserve comonotonicity, so X_0 and Y_0 are comonotone and unpriced. By (R), $\mathrm{LR}(X_0 + Y_0 + u) \in \mathbb{R}$ for every $u \neq 0$; by (P) the non-finite point of the cash line of $X_0 + Y_0$ is therefore 0, so $\Pi(X_0 + Y_0) = 0$, that is, $\Pi(X+Y) = \Pi(X) + \Pi(Y)$. Consequently $(X+Y)_0 = X_0 + Y_0$, and (1) with $s = t = u$ gives $r(X_0 + Y_0) = r(X_0) + r(Y_0)$; hence

$\rho(X+Y)=\rho(X)+\rho(Y)$. No premium is recovered from finite values, so the argument covers break-even books, pairs of which only one breaks even, the case $\mathrm{LR}\equiv 1$ on D, and pools whose premiums cancel. By Lemma S1.5(b), which needs only (P) and (N), (CM) holds, and the implied risk measure of Definition 5.12 is the functional ρ just constructed.

Converse. Assume (P), (N), (CM). Then ($\mathsf{P}_\exists$) is clear. By Proposition 5.13(a) and Lemma S1.5(a), every cash line is $\mathrm{LR}(X+c)=(\rho(X)+c)/(\Pi(X)+c)$ and ρ,Π are comonotonic additive. Let X,Y be comonotone and unpriced, so $\Pi(X)=\Pi(Y)=0$. For $s\neq 0$, $\mathrm{LR}(X+s)=(\rho(X)+s)/s\in\mathbb{R}$ and $s(\mathrm{LR}(X+s)-1)=\rho(X)$, and likewise for Y. Moreover $\Pi(X+Y)=0$, so $\mathrm{LR}(X+Y+u)\in\mathbb{R}$ for $u\neq 0$ and $u(\mathrm{LR}(X+Y+u)-1)=\rho(X+Y)=\rho(X)+\rho(Y)$, which is (1). □

Proof of Proposition 5.5. This is the first statement of Lemma S1.1. □

Lemma S1.2 (The cash offsets)**.** *Assume* ($\mathsf{P}_\exists$) *and* (R)*, and let $X\in L^\infty$, $a\in\mathbb{R}$ and $W=X-a$. Then* $\mathrm{LR}(W)\notin\mathbb{R}$ *if and only if* $a=\Pi(X)$*; and [*$\mathrm{LR}(W)=0$*, or* $\mathrm{LR}(W)\notin\mathbb{R}$ *and* $\mathrm{LR}(W+1)=1$*] if and only if* $a=\rho(X)$*. Consequently* (RC) *holds if and only if* Π *and* ρ *are uniformly continuous, and* (RM) *holds if and only if* Π *and* ρ *are monotone.*

Proof. The first statement is the definition of the price point. For the second, $\Pi(W)=\Pi(X)-a$ and $\rho(W)=\rho(X)-a$. If $\Pi(W)\neq 0$, then W is not unpriced and $\mathrm{LR}(W)=\rho(W)/\Pi(W)$, which vanishes exactly when $\rho(W)=0$. If $\Pi(W)=0$, then $\mathrm{LR}(W)\notin\mathbb{R}$ and $\mathrm{LR}(W+1)=\rho(W)+1$ by Lemma S1.1, which equals 1 exactly when $\rho(W)=0$. In both cases the condition means $a=\rho(X)$. The axioms (RC) and (RM) therefore compare $\Pi(X)$ with $\Pi(Y)$ in (i) and $\rho(X)$ with $\rho(Y)$ in (ii). □

Lemma S1.3 (Implied premium)**.** *Assume* (P)*. Then $\Pi(X+c)=\Pi(X)+c$ for all X and c, and $D=\{\Pi\neq 0\}$. If also* (N)*, then $\Pi(c)=c$ for every constant c.*

Proof. The cash line of $X+c$ is $d\mapsto\mathrm{LR}(X+c+d)$, the cash line of X shifted by c; its price point is therefore the price point of X minus c, which is (4). $X\in D$ means that 0 is not the price point, i.e. $\Pi(X)\neq 0$. Under (N) the cash line of 0 takes the value 1 at every $d\neq 0$, so its price point is 0, $\Pi(0)=0$, and $\Pi(c)=c$. □

Lemma S1.4 (Implied risk measure)**.** *Assume* (P)*,* (N)*,* (CM)*. If $X\in D$ and $X+c\in D$, then $\rho(X+c)=\rho(X)+c$. Consequently $\rho(X):=\rho(X+c)-c$, for any $c\neq-\Pi(X)$, defines a cash-additive functional on L^∞ that agrees with $\Pi\cdot\mathrm{LR}$ on D, satisfies $\rho(c)=c$, and*

$$\mathrm{LR}(X+c)=\frac{\rho(X)+c}{\Pi(X)+c}\qquad\text{for all } c\neq-\Pi(X). \tag{1}$$

Proof. Let $X,X+c\in D$ with $c\neq 0$. The constant c lies in D (by (N)), is comonotone with X, and $\Pi(X)+\Pi(c)=\Pi(X+c)\neq 0$ by Lemma S1.3. So (CM) applies to the pair (X,c) and gives $\mathrm{LR}(X+c)=(\Pi(X)\mathrm{LR}(X)+c)/(\Pi(X)+c)$, i.e. $\rho(X+c)=\Pi(X+c)\mathrm{LR}(X+c)=\rho(X)+c$.

For arbitrary X, $X+c\in D$ exactly when $c\neq-\Pi(X)$. If c,c' are two such values, then $X+c'=(X+c)+(c'-c)$ with both $X+c$ and $X+c'$ in D, so $\rho(X+c')=\rho(X+c)+(c'-c)$; hence $\rho(X+c)-c$ does not depend on c, and for $X\in D$ (take $c=0$) it equals $\Pi(X)\mathrm{LR}(X)$. Cash additivity of the extension is immediate. For $c\neq-\Pi(X)$, $\mathrm{LR}(X+c)=\rho(X+c)/\Pi(X+c)=(\rho(X)+c)/(\Pi(X)+c)$. Finally $\rho(c)=\Pi(c)\mathrm{LR}(c)=c$ for $c\neq 0$, and $\rho(0)=\rho(1)-1=0$. □

Proof of Proposition 5.13. (a) is (1), and (b) restates it. For (c), $(c+\Pi(X))\mathrm{LR}(X+c)=\rho(X)+c\to\rho(X)-\Pi(X)$ as $c\to-\Pi(X)$. For (d), the right-hand side of (1) vanishes only at $c=-\rho(X)$, which differs from the price point when $\rho(X)\neq\Pi(X)$; when $\rho(X)=\Pi(X)$ it equals 1 for every $c\neq-\Pi(X)$. Uniqueness of the extension is clear, since any cash-additive extension must satisfy (b). □

Lemma S1.5 (Comonotonic additivity)**.** *Assume* (P) *and* (N)*.*

(a) If (CM) *holds, then* ρ *and* Π *are comonotonic additive.*
(b) Conversely, if Π *is comonotonic additive and there is a comonotonic additive* $\tilde{\rho}$ *with* $\tilde{\rho} = \Pi \cdot \mathrm{LR}$ *on* D*, then* (CM) *holds and the implied risk measure equals* $\tilde{\rho}$*.*

Proof. (a) Let X, Y be comonotone, $Z = X + Y$, $r = \rho(X) + \rho(Y)$ and $q = \Pi(X) + \Pi(Y)$. Fix $s \neq -q$ and choose $u \notin \{-\Pi(X), s + \Pi(Y)\}$.

(1) $X + u$ and $Y + s - u$ are comonotone, and by Lemma S1.3 their premiums $\Pi(X) + u$ and $\Pi(Y) + s - u$ are non-zero. So both lie in D, and their premiums sum to $q + s \neq 0$.
(2) Apply (CM) to this pair and use $\rho = \Pi \cdot \mathrm{LR}$ on D with Lemma S1.4:

$$\mathrm{LR}(Z + s) = \frac{(\rho(X) + u) + (\rho(Y) + s - u)}{q + s} = \frac{r + s}{q + s} \in \mathbb{R}.$$

(3) Thus the cash line of Z is finite at every $s \neq -q$. By (P) it has exactly one non-finite point, so that point is $-q$, and $\Pi(Z) = q$.
(4) For $s \neq -q$, Lemma S1.4 gives $\rho(Z) + s = \rho(Z + s) = \Pi(Z + s)\mathrm{LR}(Z + s) = r + s$, so $\rho(Z) = r$.

The argument never asks whether X, Y or Z break even: (P) locates the price point even when the cash line equals 1 everywhere else.

(b) Let $X, Y \in D$ be comonotone with $q = \Pi(X) + \Pi(Y) \neq 0$. Then $\Pi(X + Y) = q \neq 0$, so $X + Y \in D$, and $\mathrm{LR}(X + Y) = \tilde{\rho}(X + Y)/q = (\tilde{\rho}(X) + \tilde{\rho}(Y))/q = (\Pi(X)\mathrm{LR}(X) + \Pi(Y)\mathrm{LR}(Y))/q$. Both $\tilde{\rho}$ and the implied risk measure are cash additive (the former by additivity with constants and $\tilde{\rho}(c) = \Pi(c)\mathrm{LR}(c) = c$) and agree on D, hence everywhere. □

Lemma S1.6 (Law invariance of the implied pair)**.** *Assume* (P), (N), (CM) *and* (LI)*. Then* ρ *and* Π *are law invariant.*

Proof. If $X \overset{d}{=} Y$ then $X + c \overset{d}{=} Y + c$ for every c, so by (LI) the cash lines of X and Y have the same finite values and the same non-finite locations. Hence they have the same price point, $\Pi(X) = \Pi(Y)$, and by Proposition 5.13(b) $\rho(X) = \rho(Y)$. □

Lemma S1.7 (Homogeneity and Lipschitz continuity)**.** *Assume* (P), (N), (CM)*, and let* $F \in \{\rho, \Pi\}$*. If* (RC) *or* (RM) *holds,* F *is positively homogeneous and Lipschitz; under* (RM) *it is* 1*-Lipschitz.*

Proof. By Lemma S1.2, ρ and Π are uniformly continuous under (RC), and monotone under (RM). Since X is comonotone with itself, $F(nX) = nF(X)$ by Lemma S1.5, so $F(qX) = qF(X)$ for rational $q > 0$. *Under* (RM)*:* $X \leq Y + \|X - Y\|$ gives $F(X) \leq F(Y) + \|X - Y\|$ by monotonicity and cash additivity, so F is 1-Lipschitz, and homogeneity extends to real $\lambda > 0$. *Under* (RC)*:* continuity gives $F(\lambda X) = \lambda F(X)$ for real $\lambda > 0$. Choose $\delta > 0$ with $|F(X') - F(Y')| \leq 1$ whenever $\|X' - Y'\| \leq \delta$, and apply it to tX, tY with $t = \delta/\|X - Y\|$; this gives $|F(X) - F(Y)| \leq \|X - Y\|/\delta$. □

Lemma S1.8 (Layer formula)**.** *Assume* (P), (N), (CM), (LI) *and* (RC) *(or* (RM)*). Define, for* $P(A) = p$*,* $h(p) := \rho(\mathbf{1}_A)$ *and* $g(p) := \Pi(\mathbf{1}_A)$*. These are well defined by Lemma S1.6, defined for every* $p \in [0, 1]$ *because the space is atomless, and* $h(0) = g(0) = 0$*,* $h(1) = g(1) = 1$*. For every chain* $C_1 \supset \cdots \supset C_m$*,* $x_0 \in \mathbb{R}$ *and* $a_i \geq 0$*,*

$$\rho\left(x_0 + \sum_i a_i \mathbf{1}_{C_i}\right) = x_0 + \sum_i a_i h(P(C_i)), \tag{2}$$

and likewise for Π *with* g*.*

Proof. The partial sum $S_{j-1} = \sum_{i<j} a_i \mathbf{1}_{C_i}$ attains its maximum on C_j, so it is comonotone with $\mathbf{1}_{C_j}$. Apply Lemma S1.5 and cash additivity successively, with $\rho(a\mathbf{1}_C) = a\rho(\mathbf{1}_C) = ah(P(C))$ by Lemma S1.7. The values at 0 and 1 come from $\rho(0) = \Pi(0) = 0$ and $\rho(1) = \Pi(1) = 1$. □

Lemma S1.9 (Bounded variation)**.** *Under the assumptions of Lemma S1.8,* $h, g \in \mathscr{H}_1$*. Under* (RM) *they are non-decreasing.*

Proof. Under (RM). For $p \le q$ choose nested $A \subset B$ with $P(A) = p$, $P(B) = q$; then $\mathbf{1}_A \le \mathbf{1}_B$ gives $h(p) \le h(q)$, and likewise for g.

Under (RC). Let L be a Lipschitz constant of ρ. Take $0 = p_0 < \cdots < p_n = 1$ and a uniform random variable U. Put $I_j = \{1 - p_j < U \le 1 - p_{j-1}\}$ and $\delta_j = h(p_j) - h(p_{j-1})$. For $\sigma \in \{0,1\}^n$ let $W = \sum_j 2(n-j)\mathbf{1}_{I_j}$ and $Y_\sigma = W + \sum_j \sigma_j \mathbf{1}_{I_j}$. The values $2(n-j)+\sigma_j$ are strictly decreasing in j, so W and Y_σ have the same upper sets $\{U > 1-p_j\}$, of probability p_j. By (2) and summation by parts, $\rho\left(\sum_j v_j \mathbf{1}_{I_j}\right) = \sum_j v_j \delta_j$ for every non-increasing $v \ge 0$. Hence $|\sum_j \sigma_j \delta_j| = |\rho(Y_\sigma) - \rho(W)| \le L\|Y_\sigma - W\| \le L$. Choosing $\sigma_j = \mathbf{1}\{\delta_j > 0\}$ and then $\sigma_j = \mathbf{1}\{\delta_j < 0\}$ gives $\sum_j |\delta_j| \le 2L$. □

Lemma S1.10 (Representation)**.** *Under the assumptions of Lemma S1.8,* $\rho = I_h$ *and* $\Pi = I_g$*.*

Proof. For a simple $X = x_0 + \sum_i a_i \mathbf{1}_{C_i}$ with $a_i \ge 0$ and $C_i = \{X \ge x_i\}$, a direct computation gives $I_h(X) = x_0 + \sum_i a_i h(P(C_i)) = \rho(X)$. Simple positions are uniformly dense in L^∞, and ρ and I_h are both Lipschitz (Lemma S1.7 and $|I_h(X) - I_h(Y)| \le \mathrm{TV}(h)\|X - Y\|$), so $\rho = I_h$. Likewise $\Pi = I_g$. □

Proof of Theorems 6.1 and 6.2. (i)⇒(ii). By Lemma S1.1, (P), (N) and (CM) hold. Lemmas S1.3–S1.10 then give $\Pi = I_g$ and $\rho = I_h$ with $h, g \in \mathscr{H}_1$, non-decreasing under (RM). On $D = \{\Pi \ne 0\} = \{I_g \ne 0\}$, $\mathrm{LR} = \rho/\Pi = I_h/I_g$, and off D the loss ratio is not finite. Under (S), at a book with $I_g(X) = 0$ the label is $\star$ exactly when the residue $\rho(X) = I_h(X)$ vanishes, which is the value of $[I_h(X) : 0]$.

(ii)⇒(i). Let LR be as in (ii). Since $I_g(X + c) = I_g(X) + c$, the cash line of X is non-finite exactly at $c = -I_g(X)$, so (P) holds and $\Pi = I_g$. (N) holds because $I_h(c) = I_g(c) = c$. Next, I_k is comonotonic additive for $k \in \mathscr{H}$: for comonotone X, Y, any two upper sets $\{X \ge a\}$, $\{Y \ge b\}$ are almost surely nested (otherwise comonotonicity fails on a $P \otimes P$-positive set); for simple comonotone X, Y the union of their upper sets is therefore a chain, giving $X = x_0 + \sum a_i \mathbf{1}_{C_i}$, $Y = y_0 + \sum b_i \mathbf{1}_{C_i}$ with $a_i, b_i \ge 0$, and additivity follows from the layer computation; for general X, Y approximate by the comonotone simple pairs $\lfloor nX \rfloor / n$, $\lfloor nY \rfloor / n$ and use the Lipschitz bound. Lemma S1.5(b) with $\tilde{\rho} = I_h$ gives (CM) and $\rho = I_h$. Hence ($\mathsf{P}_\exists$) and (R) hold by Lemma S1.1. (LI) holds because I_k depends on X only through $x \mapsto P(X > x)$. By Lemma S1.2 the two sure amounts in (RC) and (RM) are $I_g(X)$ and $I_h(X)$, so (RC) holds because I_h, I_g are Lipschitz, and (RM) when h, g are non-decreasing. □

Proof of Propositions 6.3 and 6.4. $g(p) = \Pi(\mathbf{1}_A)$ is minus the price point of $\mathbf{1}_A$, and $h(p) = \rho(\mathbf{1}_A)$ is given by Proposition 5.13(b); both are determined by LR, including when $h = g$. The identities for D and N follow from $\Pi = I_g$, $\rho = I_h$. Monotonicity of ρ is equivalent to monotonicity of h because I_h is monotone if and only if h is non-decreasing. Proposition 6.4 collects Lemmas S1.3–S1.5 and S1.7, the Lipschitz bound for I_k, and $\mathrm{LR}(\lambda X) = \lambda\rho(X)/(\lambda\Pi(X)) = \mathrm{LR}(X)$ on D (and $\lambda D = D$). □

S2 On the role of the axioms

S2.1 Structure of the result

The proof of Theorem 6.1 has three steps.

(A) *Implied pair.* Axioms ($\mathsf{P}_\exists$) and (R) give (P), (N) and (CM) (Lemma S1.1). Property (P) defines Π; (R) with a sure loss, equivalently (N) and (CM) applied with sure losses, defines ρ and gives the Möbius form of the cash lines (Lemmas S1.1, S1.3 and S1.4).

(B) *Transfer.* Given (P) and (N), (CM) holds if and only if ρ and Π are comonotonic additive (Lemma S1.5), and (LI) passes from LR to ρ and Π (Lemma S1.6).

(C) *Representation.* A law-invariant, comonotonic additive, uniformly continuous functional on an atomless L^∞ is a signed Choquet integral (Wang et al., 2020a, Thm. 1); under monotonicity it is a distortion premium (Schmeidler, 1986; Wang et al., 1997).

Step (C) is classical; we reprove it in Section S1 to keep the constants explicit and to show that no continuity is needed in the monotone case. Steps (A) and (B) are specific to loss ratios. Together, axioms ($\mathsf{P}_\exists$) and (R) state that LR is the ratio of a pair of comonotonic additive functionals that coincide with the identity on constants, and that this pair can be read off LR.

S2.2 Assumed and derived properties

Table S1 lists the properties of the representation and their source. Axiom ($\mathsf{P}_\exists$) is ordinal, and (LI), (RC), (RM) select the class of representing functions. The cardinal content of the axioms lies in (R), which contains the normalization (N) and the pooling property (CM); given (P) and (N), (CM) is equivalent to comonotonic additivity of the implied premium and risk measure, which is the axiom that produces Choquet representations in Yaari (1987), Schmeidler (1986) and Wang et al. (1997). The Choquet form of the numerator and the denominator is therefore a consequence of comonotonic additivity, as in the single-functional case.

Property	**Source**	**Status**
unique price point ((P)); Π cash additive	($\mathsf{P}_\exists$) and (R) (Lemma S1.1)	derived
$\mathrm{LR}(c)=1$ ((N)); $\Pi(c)=c$, $\rho(c)=c$	(R) (Lemma S1.1)	derived
ρ defined on L^∞; Möbius cash lines	(R) with a sure loss (Lemma S1.1)	derived
ρ,Π comonotonic additive; (CM)	(R) (Lemmas S1.1 and S1.5)	equivalent
ρ,Π law invariant	(LI) (Lemma S1.6)	assumed
ρ,Π continuous or monotone	(RC), (RM)	assumed
h,g of bounded variation	Lemma S1.9	derived
$\rho=I_h$, $\Pi=I_g$	Lemmas S1.8 and S1.10	derived
positive homogeneity, scale invariance	Lemma S1.7	derived
association	Corollary S5.1	derived
uniqueness of (h,g)	Proposition 6.3	derived

Table S1: Assumed and derived properties in Theorem 6.1.

S2.3 Relation to the single-functional theory

The characterizations in the literature concern premiums and risk measures; they cannot be applied to LR directly, since a loss ratio is not additive and (R) is an identity between loss ratios. What is specific to the loss ratio is the construction of the implied pair, the derivation of additivity on all comonotone pairs, in particular on the break-even set where the weights of (CM) are not identified (Remark 5.11), uniqueness of (h,g) also when $\mathrm{LR}\equiv 1$, the impossibility of a monotone loss ratio (Proposition 5.21), the structure of the associated family (Corollary S5.1), and the version for nonnegative losses (Section S4), in which the price point is not observed and must be extrapolated.

S2.4 Loss ratios excluded and included

Ratios with a proportional loading, $\mathbb{E}/((1+\theta)\mathbb{E})$, violate (R) through (N); scenario-weighted ratios $\mathbb{E}_Q/\mathbb{E}$ violate (LI); the ratio $\max(I_{\sqrt{t}}, I_{\min(1,1.5t)})/\mathbb{E}$ violates (R), its numerator being cash additive but not comonotonic additive; and ordinal transforms such as $(\rho/\Pi)^3$ satisfy ($\mathsf{P}_\exists$), (P) and (N) but violate (R). The class characterized contains, among others, the expected loss over any distortion premium (Example 6.5), ratios of expected shortfalls (Example 6.6), value-at-risk over expected shortfall, and ratios that take values on both sides of 1.

S2.5 Further remarks on the axioms

Remark S2.1 (A weight-free form of (CM)). Property (CM) is a cardinal statement whose weights are read off the price points. Given (P), the pair (N)+(CM) is equivalent to the following two statements, which use no weights:

(M) every cash line has at most one zero, and $\mathrm{LR}(X+c) = \dfrac{c-c_X^0}{c-c_X^\infty}$ for all $c \neq c_X^\infty$;

(A) if X, Y are comonotone, then $c_{X+Y}^\infty = c_X^\infty + c_Y^\infty$ and $c_{X+Y}^0 = c_X^0 + c_Y^0$.

(M) says that every cash line is the unit-slope Möbius map determined by its own zero and price point; (A) says that comonotone pooling adds the cash offsets at the levels 0 and ∞. Indeed, (N) and (CM) imply (M) by Proposition 5.13(a) and Corollary 5.14, and (A) by Lemma S1.5(a). Conversely, (A) with $X = Y = 0$ gives $c_0^\infty = c_0^0 = 0$, so (M) gives $\mathrm{LR}(c) = c/c = 1$ for $c \neq 0$, which is (N). If $X, Y \in D$ are comonotone with $c_X^\infty + c_Y^\infty \neq 0$, then $X+Y \in D$ by (A), and (M) at $c=0$ gives $\mathrm{LR}(Z) = c_Z^0/c_Z^\infty$ for $Z \in \{X, Y, X+Y\}$; by (A), $\mathrm{LR}(X+Y) = (c_X^0 + c_Y^0)/(c_X^\infty + c_Y^\infty) = w_{X,Y}\mathrm{LR}(X) + w_{Y,X}\mathrm{LR}(Y)$, which is (CM). Unlike the formulation (CP) of Remark S2.2, (M) refers to no functional other than LR, but it still uses the points c_X^∞ and c_X^0. Axiom (R) states the content of (M) and (A) without them.

Remark S2.2 (A formulation by cash lines). One may alternatively postulate (CP): every cash line is a unit-slope Möbius map, $\mathrm{LR}(X+c) = [\rho(X)+c : \Pi(X)+c]$. Proposition 5.13(a) shows that (CP) follows from (P), (N) and (CM), the labels agreeing under (S). Conversely, (CP) implies (P) and (S), and (CP) together with (CM) implies (N): if $\pi_0 := \Pi(0) \neq 0$, (CM) applied to the constants $a = b = -\pi_0/2$ forces $\mathrm{LR}(-\pi_0) \in \mathbb{R}$, although $-\pi_0$ is the price point of 0; hence $\pi_0 = 0$, and (CM) with $X = Y = a \neq 0$ gives $(\rho(0)+2a)/(2a) = (\rho(0)+a)/a$, i.e. $\rho(0) = 0$ and $\mathrm{LR}(c) = 1$. Thus (P)+(N)+(CM)+(S) is equivalent to (CP)+(CM).

The following example illustrates Proposition 5.21.

Example S2.3. Let $h = \mathrm{id}$, $g = \sqrt{\cdot}$ and $p = q = \frac{1}{4}$ in the proof. Both families of books in Figure S2 increase in the almost sure order, but the loss ratio increases along the first and decreases along the second: adding a sure loss to an over-priced book moves the ratio towards 1 from below, whereas adding risk to a sure loss moves it away from 1.

S3 Dropping law invariance

We state the version without (LI), which covers scenario-dependent loss ratios. A *game* is a map $\nu : \mathscr{F} \to \mathbb{R}$ with $\nu(\emptyset) = 0$. It is *null-invariant* if $P(A \triangle B) = 0$ implies $\nu(A) = \nu(B)$; its *chain variation* is $\|\nu\|_{bv} = \sup \sum_j |\nu(A_j) - \nu(A_{j-1})|$ over chains $\emptyset = A_0 \subset \cdots \subset A_n = \Omega$. Let $\mathscr{V}_1$ be the null-invariant games with $\|\nu\|_{bv} < \infty$ and $\nu(\Omega) = 1$, and for $\nu \in \mathscr{V}_1$ let $C_\nu(X) = \int_0^\infty \nu(X > x)\,dx + \int_{-\infty}^0 [\nu(X > x) - 1]\,dx$. A *capacity* is a monotone game.

Theorem S3.1 (Without law invariance)**.** LR *satisfies* ($\mathsf{P}_\exists$), (R) *and* (RC) *if and only if* $\mathrm{LR} = C_{\nu_1}/C_{\nu_2}$ *where* $C_{\nu_2} \neq 0$ *and* $\mathrm{LR} \notin \mathbb{R}$ *where* $C_{\nu_2} = 0$*, for* $\nu_1, \nu_2 \in \mathscr{V}_1$*; with* (RM) *in place of* (RC)*, if and only if*

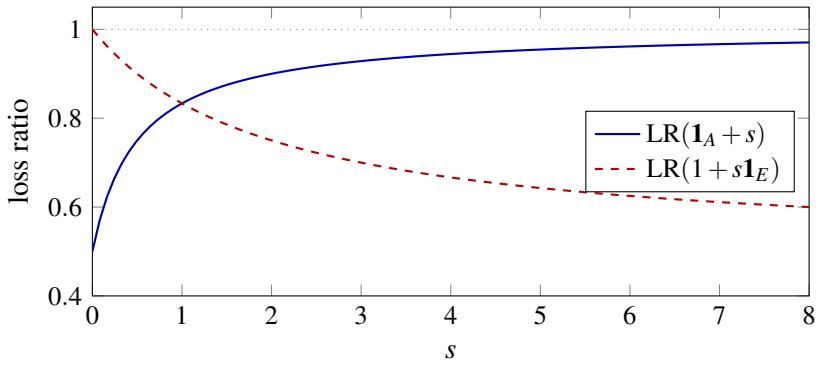


Figure S2: Example S2.3: $h = \mathrm{id}$, $g = \sqrt{\cdot}$, $P(A) = P(E) = \frac{1}{4}$. Both families of books increase with s; the loss ratio increases along one and decreases along the other.

the same holds with capacities ν_1, ν_2. The pair is unique: $\nu_1(A) = \rho(\mathbf{1}_A)$ and $\nu_2(A) = \Pi(\mathbf{1}_A)$. The loss ratio is in addition law invariant if and only if $\nu_i = k_i \circ P$ with $k_i \in \mathscr{H}_1$, which recovers Theorems 6.1 and 6.2.

The set-function representation in Theorem S3.1 is that of Murofushi et al. (1994) and Cerreia-Vioglio et al. (2012, 2015); the proof is in Section S3.1.

S3.1 Proof of Theorem S3.1

Lemmas S1.3, S1.4, S1.5 and S1.7 do not use (LI). Put $\nu_1(A) := \rho(\mathbf{1}_A)$ and $\nu_2(A) := \Pi(\mathbf{1}_A)$. They are null-invariant, because ρ and Π act on almost-sure classes, normalized by $\rho(1) = \Pi(1) = 1$, and the layer formula (2) holds with $\nu_i(C_j)$ in place of $h(P(C_j))$, for every chain $C_1 \supset \cdots \supset C_m$.

Bounded variation. Given a chain $\emptyset = A_0 \subset \cdots \subset A_n = \Omega$, run the proof of Lemma S1.9 with $I_j := A_j \setminus A_{j-1}$; then $\{W \geq 2(n-j)\} = A_j$, and one obtains $\|\nu_1\|_{bv} \leq 2L$. Under (RM), ν_1 is monotone, i.e. a capacity.

Jordan decomposition of a game. For $\nu \in \mathscr{V}$ let $\nu^+(A) := \sup \sum_j (\nu(A_j) - \nu(A_{j-1}))^+$ over chains from $\emptyset$ to A, and $\nu^- := \nu^+ - \nu$. For $A \subset B$, appending B to a chain for A gives $\nu^+(B) \geq \nu^+(A) + (\nu(B) - \nu(A))^+$. Hence ν^+ is non-decreasing, and $\nu^-(B) - \nu^-(A) \geq (\nu(B) - \nu(A))^+ - (\nu(B) - \nu(A)) \geq 0$, so ν^- is non-decreasing. Moreover $\nu^+(A) \leq \|\nu\|_{bv}$ (append Ω), and null-invariance passes to $\nu^\pm$ (if $P(A \triangle A') = 0$, map $A_j \mapsto A_j \cap A'$ for $j < n$ and $A_n \mapsto A'$, which preserves all ν-values).

Representation. $C_{\nu^\pm}$ (with the correction term by $\nu^\pm(\Omega)$) is monotone with cash slope $\nu^\pm(\Omega)$, hence Lipschitz, and $C_\nu = C_{\nu^+} - C_{\nu^-}$ by linearity in ν. Now ρ and C_{ν_1} are Lipschitz and agree on simple positions by (2), so $\rho = C_{\nu_1}$; likewise $\Pi = C_{\nu_2}$.

Converse and uniqueness. As in the proof of Theorems 6.1 and 6.2, with ν in place of $h \circ P$; null-invariance makes C_ν well defined on almost-sure classes and makes almost-sure nestedness sufficient. Uniqueness follows because $\Pi(\mathbf{1}_A)$ is minus the price point of $\mathbf{1}_A$ and $\rho(\mathbf{1}_A)$ is given by Proposition 5.13(b).

Law invariance. If (LI) holds, Lemma S1.6 makes $\nu_i(A)$ a function of $P(A)$ alone, so $\nu_i = k_i \circ P$; since the space is atomless, $\|k \circ P\|_{bv} = \mathrm{TV}(k)$, so $k_i \in \mathscr{H}_1$. Conversely $k \circ P$ gives a law-invariant functional. □

S4 Nonnegative losses

Insurance losses are nonnegative and premiums are positive. On $\mathscr{X} = L^\infty_+ \setminus \{0\}$ only upward shifts $X + c$, $c \geq 0$, are available, and the price point $-\Pi(X) < 0$ lies outside the domain, so that (P) cannot

be used; nor can ($\mathsf{P}_\exists$) and (R), which pool books after moving them to zero premium. The axioms of this section are therefore stated with reals attached to the cash lines; restating them in terms of values of LR_0 alone, as in Section 5, is left open. In this section the loss ratio is a map $\mathrm{LR}_0 : \mathscr{X} \to \mathbb{R}$, and $N_0 := \{X \in \mathscr{X} : \mathrm{LR}_0(X) = 1\}$.

S4.1 Axioms

Definition S4.1 (Axioms (A0)–(A5))**. (A0)** N_0 has empty interior in $(\mathscr{X}, \|\cdot\|_\infty)$.

(A1) For each $X \in \mathscr{X} \setminus N_0$ there are reals a_X, b_X with $\mathrm{LR}_0(X+c) = (a_X+c)/(b_X+c)$ for all $c \geq 0$. By Lemma S4.2, $a_X \neq b_X$, the pair is unique and $b_X > 0$; put $\rho(X) := a_X$ and $\Pi(X) := b_X$ on $\mathscr{X} \setminus N_0$.

(A2) LR_0 is law invariant.

(A3) ρ and Π are non-decreasing on $\mathscr{X} \setminus N_0$. Variant (A3$_s$): ρ and Π are uniformly continuous on $\mathscr{X} \setminus N_0$.

(A4) Let $\bar\rho, \bar\Pi$ be the continuous extensions of ρ, Π to $\mathscr{X}$ (they exist under (A3) or (A3$_s$) by Lemma S4.7). For comonotone $X, Y \in \mathscr{X}$:

(A4a) $\bar\Pi(X+Y) = \bar\Pi(X) + \bar\Pi(Y)$;

(A4b) $\bar\Pi(X+Y)\mathrm{LR}_0(X+Y) = \bar\Pi(X)\mathrm{LR}_0(X) + \bar\Pi(Y)\mathrm{LR}_0(Y)$.

(A5) LR_0 is norm-continuous on $\mathscr{X}$.

Axiom (A0) excludes $\mathrm{LR}_0 \equiv 1$, for which ρ and Π are not identified, and ensures that ρ, Π are defined on a dense set. Axiom (A1) is the counterpart of Proposition 5.13(a) on a half line: since the price point is not observed, the premium must be extrapolated from $c \geq 0$, which requires the functional form to be assumed rather than derived. Axiom (A4b) is the pooling identity (CM), and (A4a) states that premiums add under comonotone pooling; on $\mathscr{X}$ the two must be stated separately. Axiom (A5) excludes jumps of LR_0 where the premium distortion vanishes; it excludes, for example, value-at-risk-type premium distortions $g = \mathbf{1}_{(a,1]}$, and it implies $g > 0$ (Step C5 in Section S4.4).

Lemma S4.2. *If $X \notin N_0$ satisfies (A1) with (a,b), then $a \neq b$, the pair is unique, $b > 0$, and for every $c_0 \geq 0$ one has $X + c_0 \notin N_0$ with $\rho(X+c_0) = \rho(X) + c_0$ and $\Pi(X+c_0) = \Pi(X) + c_0$.*

S4.2 Results

Theorem S4.3. LR_0 *satisfies (A0)–(A5) if and only if* $\mathrm{LR}_0 = C_h/C_g$ *on $\mathscr{X}$ for distortion functions $h \neq g$ with $g > 0$ on $(0,1]$. In that case (h,g) is unique, also among pairs in $\mathscr{H}_1^2$ with $C_{g'} > 0$ on $\mathscr{X}$; moreover $\bar\rho = C_h$, $\bar\Pi = C_g$ and $N_0 = \mathscr{X} \cap \{C_h = C_g\}$.*

Theorem S4.4. *Axioms (A0)–(A2), (A3$_s$), (A4) and (A5) hold if and only if* $\mathrm{LR}_0 = C_h/C_g$ *with $h, g \in \mathscr{H}_1$, $h \neq g$, and $g > 0$ on $(0,1]$.*

Proposition S4.5. $\mathrm{LR}_0 = C_h/C_g$ *with distortion functions h, g and $g > 0$ on $(0,1]$ if and only if* LR_0 *is the restriction to $\mathscr{X}$ of an extended loss ratio satisfying the axioms of Theorem 6.2 with $\Pi > 0$ on $\mathscr{X}$. The representing pair (h,g) of such an extension is unique if and only if* $\mathrm{LR}_0 \not\equiv 1$. *In that case the extension is unique up to the labels $\infty, \star$ at its non-finite values, and unique under convention* (S).

Theorem S4.3 is the loss-ratio analogue of the characterization of Wang et al. (1997). Unlike that result, the pair (ρ, Π) is defined only off N_0, and positivity of g is derived from (A5). The proofs are given in Section S4.4, and Section S4.3 shows that none of (A2)–(A5) can be dropped.

Dropped	Witness on $\mathscr{X}$	Reason
(A2)	$\mathbb{E}_Q/\mathbb{E}$	N_0 lies in a closed hyperplane, so (A0) holds; not law invariant.
(A3)	$C_h/\mathbb{E}$, $h = t + \frac{1}{4}\sin 2\pi t$	$\rho(\mathbf{1}_A) > \rho(\mathbf{1}_B)$ for $\mathbf{1}_A \le \mathbf{1}_B$, $P(A) = 0.4, P(B) = 0.6$.
(A3$_s$)	$C_h/\mathbb{E}$, h continuous non-BV	Extensions exist; h is not of bounded variation.
(A4a)	$\mathbb{E}/\max(C_{\sqrt{t}}, C_{\min(1,1.5t)})$	Premium is not comonotonic additive (numbers above); (A4b) holds.
(A4b)	$\max(C_{\sqrt{t}}, C_{\min(1,1.5t)})/\mathbb{E}$	$\bar{\Pi} = \mathbb{E}$ additive, but $\bar{\Pi}\mathrm{LR}_0$ is not.
(A5)	C_h/C_g with $g = (2t-1)^+$, $h = g^2$, and $\mathrm{LR}_0 := 1$ where $C_g = 0$	LR_0 jumps: $\mathrm{LR}_0(\mathbf{1}_B + n^{-1}\mathbf{1}_C) = \frac{1}{2}$ while $\mathrm{LR}_0(\mathbf{1}_B) = 1$ for $P(B) \le \frac{1}{2}$, $C \supset B$, $P(C) = \frac{3}{4}$.

Table S2: Witnesses for (A2)–(A5) on the actuarial domain. In each case all other axioms hold.

S4.3 Witnesses for the actuarial axioms

S4.4 Proofs

Proof of Lemma S4.2. $a/b = \mathrm{LR}_0(X) \neq 1$, so $a \neq b$. *Uniqueness.* If $(a+c)(b'+c) = (a'+c)(b+c)$ for all $c \ge 0$, comparing coefficients gives $a + b' = a' + b$ and $ab' = a'b$; writing $a' = a + t$, $b' = b + t$ gives $t(a-b) = 0$, so $t = 0$. *Positivity.* If $b \le 0$, the pole $-b \ge 0$, at which the numerator does not vanish, would contradict finiteness of LR_0. *Shifts.* $\mathrm{LR}_0(X + c_0) = (a + c_0)/(b + c_0) \neq 1$, and the profile of $X + c_0$ is the shifted profile of X. □

Lemma S4.6 (Sierpiński and quantile coupling)**.** *(a) If $P(B) > 0$, $P(\cdot \mid B)$ carries a uniform random variable. (b) For every $Y \in L^\infty$ there is a uniform U with $Y = q_Y(U)$ a.s. If $X \overset{d}{=} Y$ and $X_n \to X$ in norm, then $Y_n := q_{X_n}(U)$ satisfies $Y_n \overset{d}{=} X_n$ and $\|Y_n - Y\| \le \|X_n - X\|$.*

Proof. (a) $P(\cdot \mid B)$ is atomless; by Sierpiński's theorem there are sets $B_t \subset B$, increasing in dyadic $t \in [0,1]$, with $B_1 = B$ and $P(B_t) = tP(B)$. Put $W := \inf\{t : \omega \in B_t\}$; then $\{W < t\} \subset B_t \subset \{W \le t\}$, so $P(W \le t \mid B) = t$. (b) Off the atoms of Y set $U := F_Y(Y)$; on each atom $\{Y = y\}$ set $U := F_Y(y-) + (F_Y(y) - F_Y(y-))W_y$ with W_y uniform under $P(\cdot \mid Y = y)$ by (a). Then U is uniform and $q_Y(U) = Y$. If $|X_n - X| \le \varepsilon$ a.s., quantiles move by at most ε, so $\|q_{X_n}(U) - q_X(U)\| \le \|X_n - X\|$ and $q_X = q_Y$. □

Lemma S4.7 (Extension)**.** *Assume (A0)–(A2) and (A3) (respectively (A3$_s$)). Then ρ, Π are 1-Lipschitz (respectively uniformly continuous) on $\mathscr{X} \setminus N_0$, hence extend uniquely to continuous $\bar{\rho}, \bar{\Pi}$ on $\mathscr{X}$. The extensions are cash additive for $c \ge 0$, law invariant, monotone under (A3), and $\bar{\Pi} \ge 0$.*

Proof. Let $X, Y \notin N_0$ and $t = \|X - Y\|$. Then $Y + t \notin N_0$ (Lemma S4.2) and $X \le Y + t$, so (A3) gives $\rho(X) \le \rho(Y) + t$; thus ρ, Π are 1-Lipschitz, and $\mathscr{X} \setminus N_0$ is dense by (A0). Cash additivity passes to the limit. Monotonicity: if $X \le Y$, choose $X_n, Y_n \notin N_0$ within $1/n$ of X, Y; then $X_n \le Y_n + 2/n \notin N_0$, and let $n \to \infty$. Law invariance: N_0 is law invariant by (A2), and off N_0 so are ρ, Π by uniqueness in Lemma S4.2. If $X \overset{d}{=} Y$ and $X_n \to X$ with $X_n \notin N_0$, then $Y_n := q_{X_n}(U) \notin N_0$ satisfies $\rho(Y_n) = \rho(X_n)$ and $Y_n \to Y$ (Lemma S4.6), so $\bar{\rho}(X) = \bar{\rho}(Y)$. Finally $\Pi > 0$ off N_0 gives $\bar{\Pi} \ge 0$. Under (A3$_s$) the same argument applies with extension by uniform continuity. □

Lemma S4.8 (Sweep)**.** *Let k be bounded Borel with $k(0) = 0$, such that C_k is comonotonic additive on $\mathscr{X}$ (true for $k \in \mathscr{H}$ and for continuous k). If $k(p_0) \neq 0$ for some $p_0 \in (0,1]$, then $\mathscr{X} \cap \{C_k = 0\}$ has empty interior.*

Proof. Write $X = q_X(U)$ and $A = \{U > 1 - p_0\}$. Then X and $\mathbf{1}_A$ are comonotone, so $C_k(X + t\mathbf{1}_A) = C_k(X) + t\,k(p_0)$ for $t > 0$, which vanishes for at most one t. Since $\|X + t\mathbf{1}_A - X\| = t$, every ball around X meets $\{C_k \neq 0\}$. □

Proof of Theorem S4.3. Step C1 (law invariance, extension). Lemma S4.7.

Step C2 (pole). $\bar\Pi$ is comonotonic additive by (A4a) and continuous, hence positively homogeneous. Then $\bar\Pi(2)=2\bar\Pi(1)$ and $\bar\Pi(2)=\bar\Pi(1)+1$ give $\bar\Pi(1)=1$.

Step C3 (zero). $\bar\rho$ and $\mathrm{LR}_0\bar\Pi$ are continuous (by (A5) for the second) and agree off N_0, where $a_X=\mathrm{LR}_0(X)b_X$; so $\bar\rho=\mathrm{LR}_0\bar\Pi$ on $\mathscr{X}$. By (A4b), $\bar\rho$ is comonotonic additive, hence homogeneous, and $\bar\rho(1)=1$ as in C2.

Step C4 (representation). For $P(A)=p>0$ put $h(p):=\bar\rho(\mathbf{1}_A)$, $g(p):=\bar\Pi(\mathbf{1}_A)$ and $h(0)=g(0)=0$. They are well defined (law invariance) and non-decreasing (monotonicity). Since $\varepsilon\mathbf{1}_A\le\mathbf{1}_A$, homogeneity and monotonicity give $(1-\varepsilon)h(p)\ge 0$, so $h,g\ge 0$; thus h,g are distortion functions. A simple $X\in\mathscr{X}$ equals $\sum_i(x_i-x_{i-1})\mathbf{1}_{B_i}$ with $x_{-1}=0$, $B_i=\{X\ge x_i\}$, a comonotone sum of positive multiples of indicators, so $\bar\rho(X)=C_h(X)$; approximating X by the simple $\lceil nX\rceil/n\in\mathscr{X}$ gives $\bar\rho=C_h$ on $\mathscr{X}$, and likewise $\bar\Pi=C_g$.

Step C5 (positivity of g). (a) If $g(p)=0$ then $h(p)=\mathrm{LR}_0(\mathbf{1}_A)\cdot 0=0$. (b) By (A0) some $W\notin N_0$ exists and, by Lemma S4.2, $\bar\rho(W)\ne\bar\Pi(W)$, so $h(q_0)\ne g(q_0)$ for some q_0; as $g(q_0)\ge 0$ and $g(q_0)\ne 0$ by (a), $g(q_0)>0$. (c) Suppose $g(p)=0$ for some $p>0$ and take nested B,C with $P(B)=p$, $P(C)=q_0$ and $X_n=\mathbf{1}_B+n^{-1}\mathbf{1}_C$. Then $\bar\Pi(X_n)=g(q_0)/n>0$, $\bar\rho(X_n)=h(q_0)/n$, so $\mathrm{LR}_0(X_n)=h(q_0)/g(q_0)\ne 1$. But $\bar\Pi(\mathbf{1}_B)=g(p)=0$ forces $\mathbf{1}_B\in N_0$, so $\mathrm{LR}_0(\mathbf{1}_B)=1$, contradicting (A5) since $X_n\to\mathbf{1}_B$.

Step C6 (conclusion). $\mathrm{LR}_0=\bar\rho/\bar\Pi=C_h/C_g$. If $h=g$ then $N_0=\mathscr{X}$, contradicting (A0).

Converse. Let $\mathrm{LR}_0=C_h/C_g$ with $h\ne g$ and $g>0$ on $(0,1]$; then $C_g>0$ on $\mathscr{X}$, because $P(X>x)>0$ for $x\in[0,\operatorname{ess\,sup}X)$. (A1) holds with $a_X=C_h(X)$, $b_X=C_g(X)$; (A2)–(A5) are immediate; (A0) follows from Lemma S4.8 with $k=h-g$, since $N_0=\mathscr{X}\cap\{C_{h-g}=0\}$.

Uniqueness among signed pairs. If also $\mathrm{LR}_0=C_{h'}/C_{g'}$ with $(h',g')\in\mathscr{H}_1^2$ and $C_{g'}>0$, then by Lemma S4.2 $C_{h'}=\rho$ and $C_{g'}=\Pi$ off N_0. The set N_0 is determined by LR_0 and has dense complement (Lemma S4.8, $h\ne g$), so by continuity $C_{h'}=\bar\rho=C_h$ and $C_{g'}=C_g$; evaluating at indicators gives $h'=h$, $g'=g$. □

Proof of Theorem S4.4. Steps C1, C2 and the extension hold with uniform continuity in place of monotonicity. For $Z\in L^\infty$ and $M>\|Z\|$ put $\tilde\rho(Z):=\bar\rho(Z+M)-M$, independent of M by cash additivity. Then $\tilde\rho$ is law invariant, cash additive, uniformly continuous and comonotonic additive (the shifts Z_i+M are comonotone elements of $\mathscr{X}$, and Step C3 with (A4b) applies). Lemmas S1.7–S1.10, applied to $\tilde\rho$ and likewise to $\tilde\Pi$ (using (A4a)), give $\tilde\rho=I_h$, $\tilde\Pi=I_g$ with $h,g\in\mathscr{H}_1$. Step C5 holds verbatim: in (b) the sign $g\ge 0$ comes from $\bar\Pi\ge 0$, not from monotonicity, and in (c) nested sets are comonotone in either direction. The converse uses the Lipschitz bounds for (A3$_s$) and Lemma S4.8 for (A0). □

Proof of Proposition S4.5. ($\Rightarrow$) Extend LR_0 to L^∞ by $[I_h:I_g]$; this satisfies Theorem 6.2, and $I_g>0$ on $\mathscr{X}$ because $g>0$. ($\Leftarrow$) Theorem 6.2 gives distortions h,g, and $I_g>0$ on $\mathscr{X}$ forces $g>0$ on $(0,1]$ (otherwise $I_g(\mathbf{1}_A)=g(P(A))=0$ for some A). *Uniqueness.* If $\mathrm{LR}_0\not\equiv 1$ then $h\ne g$, and two extensions have pairs that agree by the uniqueness in Theorem S4.3. Their finite values, I_h/I_g on $\{I_g\ne 0\}$, and their non-finite locations, $\{I_g=0\}$, therefore coincide; the labels at the non-finite values are fixed by (S) and are otherwise arbitrary (for instance, the value at 0 may be ∞ or $\star$). If $\mathrm{LR}_0\equiv 1$, $[I_{\mathrm{id}}:I_{\mathrm{id}}]$ and $[I_{\sqrt{\cdot}}:I_{\sqrt{\cdot}}]$ have different representing pairs: the price point of $\mathbf{1}_A$ is $-P(A)$ for the first and $-\sqrt{P(A)}$ for the second. □

S5 Consequences

Throughout this section the axioms of Theorem 6.1 and the labelling convention (S) hold, $\mathrm{LR}=[I_h:I_g]$, and for $\beta\ne 1$ we put

$$\rho_\beta:=\frac{\rho-\beta\Pi}{1-\beta},\qquad k_\beta:=\frac{h-\beta g}{1-\beta}=g+\frac{h-g}{1-\beta}\in\mathscr{H}_1.$$

Corollary S5.1 (Association). *(a)* $\rho_\beta = I_{k_\beta}$ *is cash additive of slope* 1*, and* $\rho_\beta(X) = 0 \iff \mathrm{LR}(X) \in \{\beta, \star\}$*. Hence* $\mathrm{LR}(X) = \beta \iff \rho_\beta(X) = 0$ *holds exactly off* $\{\rho = \Pi = 0\} = \{\mathrm{LR} = \star\}$*; it fails, for instance, at* $X = 0$ *and at* $X - \rho(X)$ *for every* $X \in N$*.*

(b) For $X \notin N$*:* $\mathrm{LR}(X - m) = \beta \iff m = \rho_\beta(X)$*, and then* $\Pi(X) - m = (\Pi(X) - \rho(X))/(1-\beta) \neq 0$*.*

(c) For $X \in N$*:* $\mathrm{LR}(X - m) \in \{1, \star\}$ *for every* m*.*

(d) ρ_β *is* $\mathrm{TV}(k_\beta)$*-Lipschitz; in Theorem 6.2,* $\mathrm{TV}(k_\beta) \leq (1 + |\beta|)/|1 - \beta|$*.*

(e) ρ_β *is monotone iff* k_β *is non-decreasing. If* $h \neq g$*,* ρ_β *is* not *monotone for all* β *in a punctured neighbourhood of* 1*.*

Proof. (a) $\rho_\beta(X) = 0 \iff \rho(X) = \beta\Pi(X)$; if $\Pi(X) \neq 0$ this says $\mathrm{LR}(X) = \beta$, and if $\Pi(X) = 0$ it says $\rho(X) = 0$, i.e. $\mathrm{LR}(X) = \star$. Conversely $\mathrm{LR}(X) = \star$ means $\rho = \Pi = 0$. At $X \in N$, $Y = X - \rho(X)$ has $\rho(Y) = \Pi(Y) = 0$, so $\mathrm{LR}(Y) = \star \neq \beta$ while $\rho_\beta(Y) = 0$. (b) $\mathrm{LR}(X - m) = \beta \iff \rho - m = \beta(\Pi - m) \iff m = \rho_\beta(X)$, and $\Pi - \rho_\beta = (\Pi - \rho)/(1-\beta) \neq 0$ off N. (c) On N the cash line equals 1 away from its price point (Proposition 5.13(d)). (d) Lemma S1.7 and the Lipschitz bound for I_k. (e) I_k is monotone iff k is non-decreasing. Since $h - g$ vanishes at 0 and 1 and is non-constant, there are $p < q$ with $(h-g)(p) > (h-g)(q)$ and $p' < q'$ with $(h-g)(p') < (h-g)(q')$. As $\beta \to 1$, $(1-\beta)k_\beta = h - \beta g \to h - g$ uniformly, so for β near 1 both strict inequalities persist for k_β, whatever the sign of $1 - \beta$. □

Part (e) is the counterpart of Proposition 5.21 for the associated family: even when ρ and Π are distortion premia, ρ_β is in general a signed Choquet integral.

Example S5.2. Let $h = \mathrm{id}$ and $g = \sqrt{\cdot}$. Then $k_\beta(p) = (p - \beta\sqrt{p})/(1-\beta)$ and $k'_\beta(p) = \big(1 - \beta/(2\sqrt{p})\big)/(1-\beta)$. For $\beta < 1$, k_β is non-decreasing if and only if $\beta \leq 2\sqrt{p}$ for all $p \in (0,1]$, that is, $\beta \leq 0$; for $\beta > 1$ it is non-decreasing if and only if $\beta \geq 2\sqrt{p}$ for all p, that is, $\beta \geq 2$. Hence ρ_β is a distortion premium if and only if $\beta \notin (0,2)$; for $\beta \in (0,1) \cup (1,2)$ it is a signed Choquet integral that is not monotone (Figure S3).

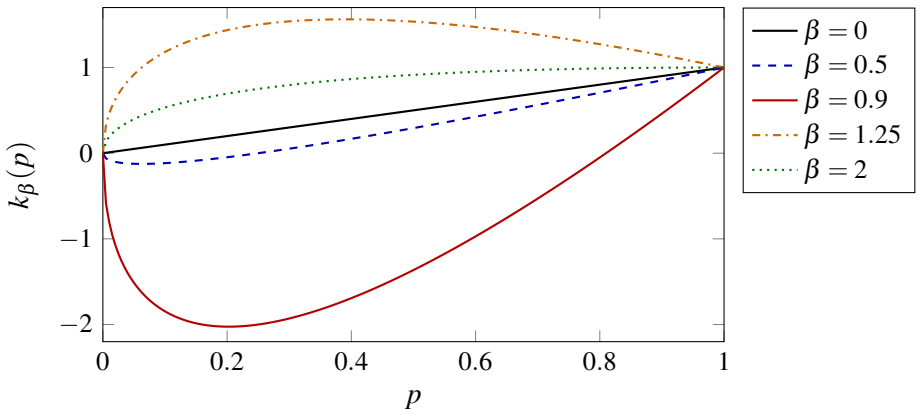


Figure S3: The distortions k_β of the associated family in Example S5.2 ($h = \mathrm{id}$, $g = \sqrt{\cdot}$). They are monotone for $\beta = 0$ and $\beta = 2$ and not monotone for $\beta \in (0,1) \cup (1,2)$.

Corollary S5.3 (Pooling at non-finite values)**.** *Assume the axioms of Theorem 6.1 and* (S)*. Then for all comonotone* X, Y*,*

$$\mathrm{LR}(X + Y) = [\rho(X) + \rho(Y) : \Pi(X) + \Pi(Y)].$$

In terms of loss ratios and implied premiums alone:

Case	*Pooled loss ratio*
$X, Y \in D$, $\Pi(X)+\Pi(Y) \neq 0$	*relative-cost-weighted combination* ((CM))
$X, Y \in D$, *premiums cancel*	∞ *if* $\mathrm{LR}(X) \neq \mathrm{LR}(Y)$; $\star$ *if* $\mathrm{LR}(X) = \mathrm{LR}(Y)$
$\Pi(X) = 0$, $Y \in D$	$\mathrm{LR}(Y) + \rho(X)/\Pi(Y)$, *with* $\rho(X) = \lim_{c\to 0} c\,\mathrm{LR}(X+c)$
$\Pi(X) = \Pi(Y) = 0$	∞ *unless* $\rho(X) = -\rho(Y)$, *in which case* $\star$

Proof. ρ and Π are comonotonic additive (Lemma S1.5) and $\mathrm{LR} = [\rho : \Pi]$ under (S) (Theorem 6.1). The table spells out $[\rho(X)+\rho(Y) : \Pi(X)+\Pi(Y)]$, using $\rho = \Pi\mathrm{LR}$ on D and Proposition 5.13(c) for books with zero premium. If premiums cancel, $\rho(X)+\rho(Y) = \Pi(X)(\mathrm{LR}(X) - \mathrm{LR}(Y))$. □

The third row has a direct reading: a loss that is carried without premium adds its size, relative to the premium base, to the loss ratio of the book it joins.

Corollary S5.4 (Orderings, convexity and shape within the class)**.** *(a)* Endpoints. ρ *is monotone iff* h *is non-decreasing;* ρ *is convex iff* h *is concave (Wang et al., 2020a, Thm. 3);* ρ *is coherent iff* h *is concave and non-decreasing, in which case* ρ *is a mixture of expected shortfalls* ES_α *over levels* $\alpha \in [0,1]$*, with the convention* $\mathrm{ES}_1 := \operatorname{ess\,sup}$ *(Kusuoka, 2001; Jouini et al., 2006; Föllmer and Schied, 2016) and Wang et al. (2020b, Thm. 5); the endpoint is needed, for instance, for* $h = \mathbf{1}_{(0,1]}$*, which gives* $\rho = \operatorname{ess\,sup}$*. The same holds for* Π *and* g*.*

(b) Ordering. $\mathrm{LR} \le 1$ *on* $\{\Pi > 0\}$ *iff* $h \le g$*.*

(c) Convex acceptance cones. *For* $\beta \neq 1$ *let* $A_\beta := \{\rho \le \beta\Pi\} = \{I_{h-\beta g} \le 0\}$*. Then* A_β *is convex iff* $h - \beta g$ *is concave, and all* A_β*,* $\beta \in [0,\infty)\setminus\{1\}$*, are convex iff* h *is concave and* g *is convex.*

(d) Quasi-convexity and quasi-concavity (sufficient conditions). *If* h *is concave and* g *is convex, then* $\{\Pi > 0\}$ *is convex,* $\mathrm{LR} \ge 1$ *there, and* LR *is quasi-convex on it. If, in Theorem 6.2 with* $g > 0$*,* h *is convex and* g *is concave, then* $\mathrm{LR} \le 1$ *on* $\mathscr{X}$ *and* LR *is quasi-concave on* $\mathscr{X}$*. In particular the actuarial ratio* $\mathbb{E}/C_g$ *with* g *concave is quasi-*concave.

(e) Rigidity. ρ_β *is convex for all* $\beta \in [0,\infty)\setminus\{1\}$ *iff* $h = g$ *and* g *is concave.*

Proof. (a) The monotonicity criterion is in Wang et al. (2020a), see also Wang et al. (2020b, Prop. 2), and the convexity criterion is Wang et al. (2020a, Thm. 3). (b) If $h \le g$ then $g - h \ge 0$ with $(g-h)(1) = 0$, so $I_{g-h} \ge 0$ and $\mathrm{LR} \le 1$ where $\Pi > 0$. If $h(p) > g(p)$ for some p, then $\mathrm{LR}(\mathbf{1}_A + c) = (h(p)+c)/(g(p)+c) > 1$ for large c, where the premium $g(p)+c$ is positive. (c) $I_{h-\beta g}$ has cash slope $1-\beta \neq 0$, so all its sublevel sets are translates of A_β; hence convexity of A_β is quasi-convexity of $I_{h-\beta g}$, which is equivalent to concavity of $h - \beta g$ (Wang et al., 2020a, Thm. 3). Taking $\beta = 0$ and $\beta \to \infty$ in $(h-\beta g)/\beta$ gives the second statement. (d) If h is concave and g is convex then I_g is concave, so $\{I_g > 0\}$ is convex; $h \ge \mathrm{id} \ge g$, so $\mathrm{LR} \ge 1$; and for $\beta \ge 1$ the set $\{I_h - \beta I_g \le 0\}$ is convex because $h - \beta g$ is concave. If h is convex and g concave, then $h \le \mathrm{id} \le g$, and $I_h - tI_g$ is concave for $t \ge 0$, so $\{\mathrm{LR} \ge t\}$ is convex; for $t < 0$ the superlevel set is all of $\mathscr{X}$, since $I_h \ge 0$ and $I_g > 0$ on $\mathscr{X}$. (e) Convexity of $\rho_\beta = I_{k_\beta}$ for $\beta < 1$ means $h - \beta g$ concave, and for $\beta > 1$ means $h - \beta g$ convex. Letting $\beta \to 1^{\mp}$ gives $h - g$ concave and convex, hence affine with zero endpoints, i.e. $h = g$; then $k_\beta = g$ and ρ_β is convex iff g is concave. The converse is clear. □

In particular, the actuarial ratio $\mathbb{E}/C_g$ with g concave is quasi-concave and not, in general, quasi-convex.

Example S5.5. Let A, B be disjoint with $P(A) = 0.2$, $P(B) = 0.3$, and $X_t = (1-t)\mathbf{1}_A + t\mathbf{1}_B$, $t \in [0,1]$. For $\mathrm{ES}_{0.9}/\mathbb{E}$ (Example 6.6), $\mathrm{ES}_{0.9}(X_t) = \max\{1-t, t\}$ and $\mathrm{LR}(X_t) = \max\{1-t,t\}/(0.2+0.1t)$, which equals 5 at $t = 0$, $10/3$ at $t = 1$ and 2 at $t = \frac{1}{2}$: the ratio lies below the larger endpoint value, as quasi-convexity requires. For $\mathbb{E}/C_{\sqrt{\cdot}}$ the ratio equals $\sqrt{0.2} \approx 0.447$ at $t = 0$, $\sqrt{0.3} \approx 0.548$ at $t = 1$ and $\sqrt{0.5} \approx 0.707$ at $t = \frac{1}{2}$: it lies above the smaller endpoint value, as quasi-concavity requires (Figure S4).

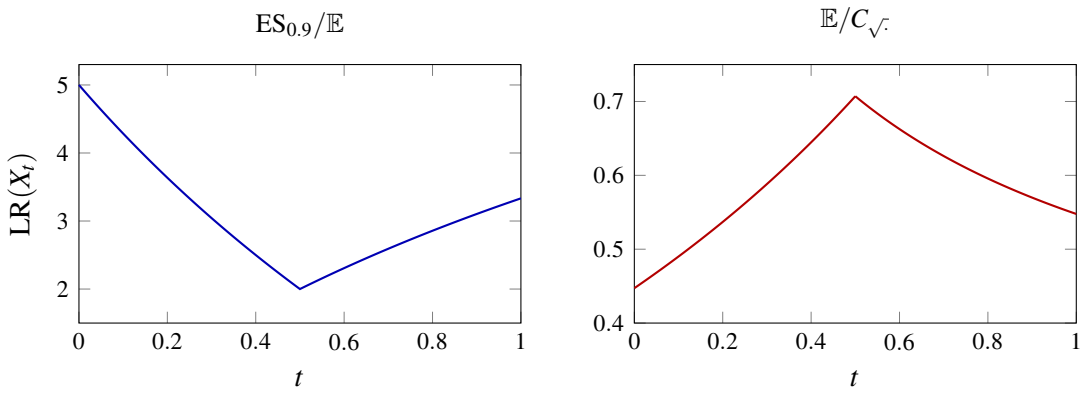


Figure S4: Loss ratios along the segment $X_t = (1-t)\mathbf{1}_A + t\mathbf{1}_B$ of Example S5.5: quasi-convex for $\mathrm{ES}_{0.9}/\mathbb{E}$ (left) and quasi-concave for $\mathbb{E}/C_{\sqrt{\cdot}}$ (right).

Example S5.6 (Further special cases)**.** (1) *Wang–Young–Panjer pairs* on nonnegative losses: Theorem S4.3. (2) *Coherent pairs*: h, g concave non-decreasing; ρ_β is a signed mixture of expected shortfalls. (3) *Actuarial ratio* $\mathbb{E}/C_g$, $h = \mathrm{id}$: Proposition 6.3 gives $g(p) = p/\mathrm{LR}(\mathbf{1}_A)$ for $P(A) = p$, so the premium distortion is identified from the loss ratios of Bernoulli books. (4) *VaR-type pairs* are admissible in Theorem 6.2; ρ_β is non-monotone near $\beta = 1$; a VaR-type premium $g = \mathbf{1}_{(a,1]}$ is excluded from Theorem S4.3, since $g > 0$ fails. (5) *Gini-type ratios*: $\mathrm{LR} - 1 = I_{h-g}/I_g$ is a location-invariant signed Choquet "variability" divided by a premium (compare Bellini et al., 2022).

S6 Positioning in the literature

Table S3 summarizes how the case-study result relates to the literatures reviewed in Section 3 of the paper.

S7 The Lean check: process record

The Lean file was written by the main assistant. Lean could not be installed in the assistant's cloud workspace, whose network policy blocks the servers that distribute Lean, so the file was compiled on the author's computer. The author installed Lean and created the project; the assistant read the build output from the project folder through the link between the Claude desktop application and the author's computer, wrote corrected versions of the file into the same folder, and the author ran the next build (Figure S5).

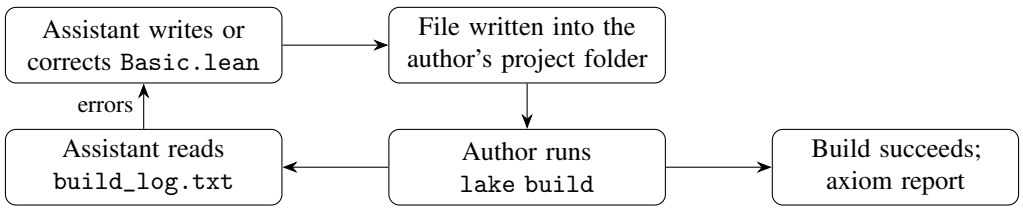


Figure S5: The build-and-repair loop used for the Lean check.

Table S4 gives the timeline. Times are in UTC and are taken from the session records and from the modification times of the files in the project folder.

Literature	What it characterizes	Difference from this paper
Yaari (1987); Wang, Young, Panjer (1997)	One premium functional H; axioms on H (law invariance, monotonicity, comonotonic additivity).	We characterize a *ratio* of two such functionals with axioms on the ratio. Monotonicity is placed on the *implied premium and risk measure*, never on the ratio.
Schmeidler (1986); Murofushi–Sugeno–Machida (1994); Cerreia-Vioglio et al. (2012, 2015); Wang–Wei–Willmot (2020)	Representation of one comonotonic additive functional as a (signed) Choquet integral.	Used as the engine (applied to ρ and Π). New: the axioms are on LR, comonotonic additivity of ρ and Π is *derived* (also on the break-even set), and uniqueness is unconditional.
Cherny–Madan (2009); Bielecki–Cialenco–Zhang (2014); Biagini–Bion-Nadal (2014)	Monotone, quasi-concave indices $\Leftrightarrow$ increasing families of coherent measures.	LR cannot be monotone (Prop. 5.21); the associated family is signed and non-monotone. Cardinal (pooling identity) rather than ordinal.
Cheridito–Kromer (2013)	Axioms for reward–risk ratios (sufficient conditions).	Reward over risk is monotone; loss over premium is not. We give an *iff*, within a stated class.
Han–Lin–Wang (2025)	Diversification quotient: iff axiomatization of a level-crossing index.	Closest in spirit; monotone families and a portfolio index. We characterize the ratio and its parts.
Haezendonck–Goovaerts (1982); Li–Wang (2023); Bellini et al. (2018)	Implicitly defined premia/risk measures, return risk measures.	Existence and uniqueness of an implicit functional, not a ratio representation. The association is recovered as a corollary and shown to hold only off $\{\rho = \Pi = 0\}$.
Chew (1983); Dekel (1986)	Ratio (weighted utility) versus implicit (betweenness) representations.	Explains why a mediant inequality is not enough and a premium-weighted identity is needed.
Gneiting (2011)	Elicitability of ratios of expectations.	Statistical, no axioms on pooling or cash shifts.

Table S3: Positioning of the case-study result. "Iff" means a characterization by necessary and sufficient axioms.

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Simone Cerreia-Vioglio, Fabio Maccheroni, Massimo Marinacci, and Luigi Montrucchio. Signed integral representations of comonotonic additive functionals. *Journal of Mathematical Analysis and Applications*, 385:895–912, 2012.

Simone Cerreia-Vioglio, Fabio Maccheroni, Massimo Marinacci, and Luigi Montrucchio. Choquet integration on Riesz spaces and dual comonotonicity. *Transactions of the American Mathematical Society*, 367, 2015.

Hans Föllmer and Alexander Schied. *Stochastic Finance: An Introduction in Discrete Time*. De Gruyter, Berlin, 4th edition, 2016.

Time (UTC)	Step
21 Oct, 21:38	The assistant writes a first Lean file for the results in Table 6; it cannot be compiled in the cloud workspace.
24 Oct, 17:08	The author asks how a formal proof check can be used; the assistant describes what is feasible (the loss-ratio steps), what is hard (the representation step) and how to install Lean.
24 Oct, 17:26–17:31	The author installs the Lean extension for the code editor and the Lean toolchain manager.
24 Oct, 17:35–17:37	The author creates the project `loss_ratio_lean` with Mathlib; Lean and Mathlib version 4.36.0-rc1; the precompiled Mathlib files are downloaded.
24 Oct, 17:38–17:39	The assistant inspects the project configuration, confirms that it is complete, and places the Lean file in it, replacing the template placeholder.
25 Oct, 12:47–12:53	First attempt to build: the command `lake` is not found, because the toolchain had not been added to the terminal's search path; fixed by loading the toolchain environment.
25 Oct, 12:55	Build 1: four errors, none in a definition or statement. One proof stopped one algebraic step short; in three proofs Lean's simplifier rewrote $\mathrm{const}(-c)$ as $-\mathrm{const}(c)$, so that cash additivity no longer applied.
25 Oct, 12:57	The assistant writes a corrected file and appends `#print axioms` for ten main theorems.
25 Oct, 13:04	Build 2: one error, in the converse construction, where the goal was displayed in an unexpanded form. The axiom report shows the three standard axioms for nine theorems and `sorryAx` for the converse.
25 Oct, 13:05	The assistant adds one line that states the goal explicitly.
25 Oct, 13:08	Build 3 succeeds. All ten theorems depend only on the three standard axioms.
25 Oct, 13:09	The assistant compares the Lean definitions of (P), (N), (CM) and (LI) and the theorem statements with Section 5.
25 Oct, 13:37	At the author's request, the project's own build output is deleted and the file is rebuilt from scratch: success in 8.8 s, with the same axiom report.

Table S4: Timeline of the Lean check.

Elyes Jouini, Walter Schachermayer, and Nizar Touzi. Law invariant risk measures have the Fatou property. *Advances in Mathematical Economics*, 9:49–71, 2006.

Shigeo Kusuoka. On law invariant coherent risk measures. *Advances in Mathematical Economics*, 3: 83–95, 2001.

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David Schmeidler. Integral representation without additivity. *Proceedings of the American Mathematical Society*, 97(2):255–261, 1986.

Qiuqi Wang, Ruodu Wang, and Yunran Wei. Distortion riskmetrics on general spaces. *ASTIN Bulletin*, 50(3):827–851, 2020b.

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